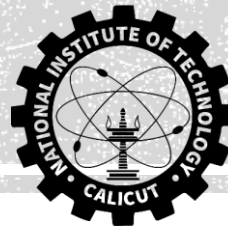


QUALITY, RELIABILITY AND MAINTENANCE

ME327E



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What is Acceptance Sampling?

■ **Definition:**

- Acceptance sampling is a statistical quality control technique used to determine whether to accept or reject a batch (lot) of products by inspecting only a sample of items, not the entire lot.

■ **Attribute vs. Variable Inspection:**

- Attribute: Classifies each item as conforming or nonconforming (does not specify how much it deviates).
- Variable: Measures a numerical quality characteristic (e.g., weight, length, concentration).

Acceptance sampling can be performed during inspection of incoming raw materials, components, and assemblies, in various phases of in-process operations, or during final product or service inspection.

Purpose of Acceptance Sampling

Key Purpose:

To decide whether to accept or reject a batch of products based on the inspection results from a sample rather than 100% inspection.

Why Not Inspect Everything?

- Destructive testing may make 100% inspection impossible.
- Sampling saves time & resources.
- Reduces inspection errors due to fatigue.

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Where is Acceptance Sampling Used?



Common Applications:

- Incoming materials: Quality assurance for parts/raw materials.
- In-process inspection: Quality checks during manufacturing.
- Finished products: Final inspection before sale or shipment.
- Inter-organizational: Between companies and suppliers/vendors.
- Internal checks: Between departments or divisions within a company

Why is Acceptance Sampling Relevant?

- Sampling doesn't improve process quality—only decides batch acceptance.
- Quality must be designed and built into products/processes.
- Acceptance sampling helps monitor but cannot "inspect quality into" products.

Advantages and Disadvantages of Sampling

Advantages

- Essential where inspection is destructive
- Economical and less handling damage
- Reduces inspection errors (less fatigue)
- Motivates suppliers/processes to improve quality

Disadvantages

- Risk of accepting bad or rejecting good lots (risks)
- Less information than 100% inspection
- More planning/documentation than 100% inspection

Producer's Risk & Consumer's Risk in Acceptance Sampling

Producer's Risk (α):

The probability of rejecting a "good" lot, typically one with an acceptable low level of nonconformance. This risk arises due to the random nature of sampling.

Goal: Minimize α to avoid unnecessary rejection of acceptable lots.

Consumer's Risk (β):

The probability of accepting a "bad" lot, one with a high or unacceptable level of nonconformance. This risk also stems from sampling variation.

Goal: Minimize β to protect the consumer from poor quality.

Quality Levels - AQL & LQL

Term	Description	Interpretation
AQL (Acceptable Quality Level)	Maximum % nonconforming considered "good"	Sets the threshold for what producers want accepted
LQL/RQL (Limiting/Rejectable Quality Level)	Minimum % nonconforming considered "bad"	Sets the threshold for what consumers want rejected

Operating Characteristic Curve

- The Operating Characteristic (OC) curve is a graphical representation of the performance of a sampling plan.
- Plots the probability of accepting a lot (P_a) (y-axis) against the proportion nonconforming (p) in the lot (x-axis).
- Discriminatory Power:
Shows how well a sampling plan can distinguish between “good” (low defect) and “bad” (high defect) lots.
- Decision Rule:
 - High P_a for good quality.
 - Low P_a for poor quality.
- Sampling Parameters Impact:
Sample size (n), Acceptance number (c), Lot size (N) affect the curve shape.

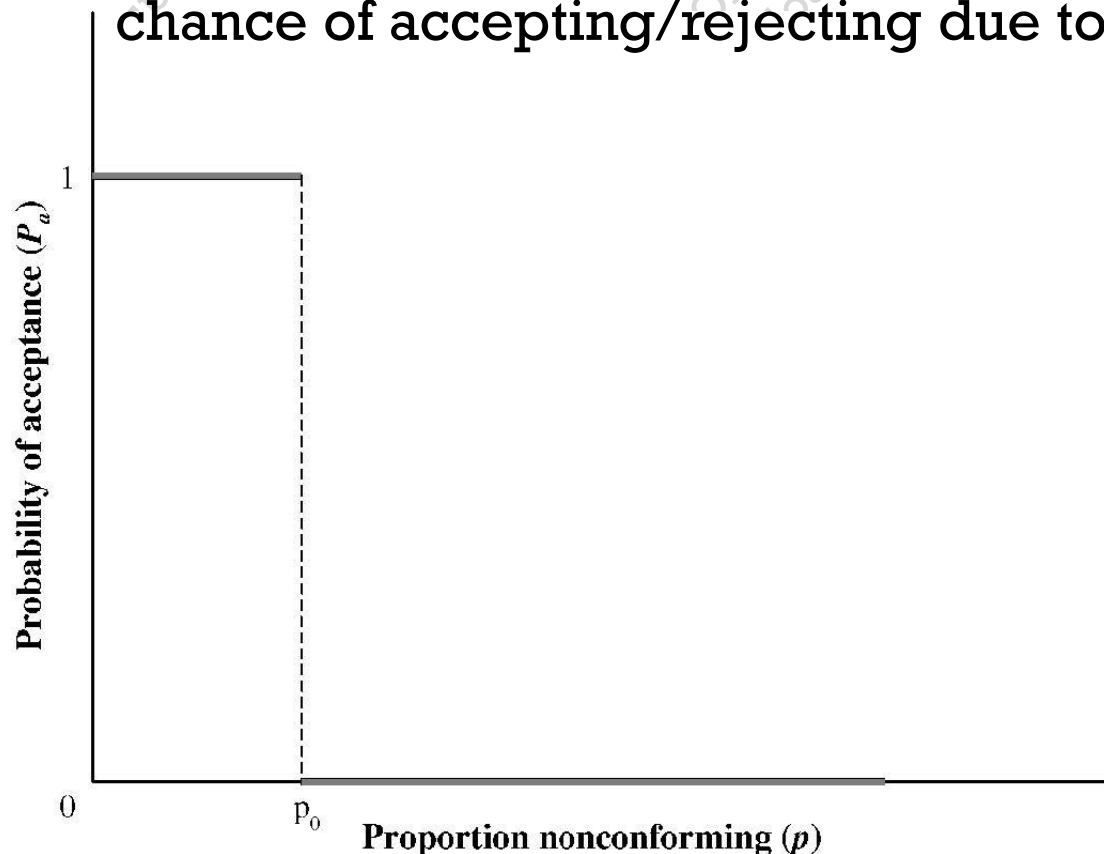
The Ideal OC Curve (Conceptual)

Ideal Shape:

- Probability of acceptance (P_a) is 1 for all lots with defect proportion $\leq p_0$ (all “good” lots accepted).
- Probability drops instantly to 0 for $p > p_0$ (all “bad” lots rejected).

Reality:

- Practical OC curves are not ideal—in practice, there's always a chance of accepting/rejecting due to sampling variation.



Operating Characteristic Curve

- In practice, however, the shape of the OC curve is not ideal.
- If the observed number of nonconforming items or nonconformities is less than or equal to c , the lot is accepted. Otherwise, the lot is rejected.
- To construct a type A OC curve, we assume that the sample is chosen from an isolated lot of finite size.
- The probability of accepting the lot is calculated based on a **hypergeometric distribution**.
- The probability of finding x nonconforming items in the sample is given by, $P(x) = \frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}}$

Where D represents the number of nonconforming items in the lot. Since the lot will be accepted if c or fewer nonconforming items are found, the probability of lot acceptance is

$$P_a = P(x \leq c) = \sum_{x=0}^c P(x)$$

Operating Characteristic Curve

- To construct a type B OC curve, we assume that a stream of lots is produced by the process and that the lot size is large (atleast 10 times) compared to the sample size.
- A binomial distribution can be used to find the probability of observing x nonconforming items in a sample of size n .
- Assuming the lot proportion nonconforming is p , this probability is given by,
$$P(x) = \binom{n}{x} p^x (1 - p)^{n-x}$$
- If the lot size is large and the probability of a nonconforming item is small
- Poisson distribution can be used as an approximation to the binomial distribution.
- The probability of x nonconforming items in the sample is found from,
$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Where, $\lambda = np$ represents the average number of nonconforming items in the sample